MIL-C-24643/32B
3 October 1986
SUPERSEDING
MIL-C-0024643/32A(SH)
8 November 1985
MIL-C-24643/32
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SWAU (INCLUDING VARIATION LS2SWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS2SWAU (unarmored) and LS2SWA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, in accordance with ASTM B 286, size 22-7, uncoated. Maximum diameter 0.033 inch.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Cross-linked polyethylene insulation. Nominal thickness 0.013 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.

Fourth - Two conductors, one black and one white, cabled together with a nominal lay of 2.5 inches to form a pair.

Fifth - Binder tape over each pair, at manufacturer's option, applied helically with overlap.

Sixth - Braided shield of AWG no. 34 or no. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.

Seventh - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.

(B) denotes changes.

AMSC N/A FSC 6145
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Eighth - The required number of pairs (see table I) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers shall be employed to form a firm, well rounded assembly.

Ninth - Binder tape applied helically with overlap.

Construction variant for type LS2SWAU

Tenth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LS2SWA

Tenth - Cross-linked polyolefin jacket. (See table I for thickness).

Eleventh - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/32	Size 2/	Number of pairs	Cable jacket thickness minimum (inch)	Overall diameter (inches)			
				Type LS2SWAU		Type LS2SWA	
				min	max	min	max
-01UN	3	3	0.040	0.480	0.520	0.530	0.570
-02UN	7	7	.040	.610	.660	.660	.710
-03UN	10	10	.040	.770	.830	.820	.880
-04UN	14	14	.050	.860	.930	.910	.980
-05UN	19	19	.050	.970	1.040	1.020	1.090
-06UN	24	24	.050	1.120	1.210	1.170	1.260
-07UN	30	30	.050	1.190	1.280	1.240	1.330
-08UN	37	37	.050	1.290	1.380	1.340	1.430
-09UN	44	44	.050	1.460	1.550	1.510	1.600
-10UN	61	61	.060	1.660	1.740	1.710	1.790

1/ Unarmored. When armor option is required, see MIL-C-24643 configuration of part number.

2/ Type LS2SWAU and LS2SWA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.37
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Shield to shield.....	500
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) - leakage at 300 lb/in^2, 2 hours, in^3, maximum	1.0
Capacitance	
Mutual capacitance - at 1 MHz, pF/ft, maximum	30
Capacitance unbalance - percent, maximum	8
Characteristic impedance - at 1 MHz, ohms (one test for each 5000 feet or fraction thereof)	75 ± 5
Group B:	
Attenuation - at 3 MHz, decibels per 100 feet, maximum	3
Drip - 75 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Pulse response time - nanoseconds/500 feet, maximum	400
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - (for type LS2SWAU only) cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Armor - conformance to material, construction and coverage (type LS2SWA only)	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 30	1000
37 through 61	500

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-32)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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